Performance Prediction and Optimization of Supercapacitors Through Data-Driven Analysis

A Project Report

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DECLARATION OF ORIGINALITY

I, **Ashish Pratap Singh**, with Roll No: **EC21B1092** hereby declare that the material presented in the Project Report titled **Performance Prediction and Optimization of Supercapacitors Through Data-Driven Analysis** represents original work carried out by me in the **Department of Electronics and Communication Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.
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Ashish Pratap Singh

Ashish Pratap Singh

Place: Chennai
Date: 09.05.2025

CERTIFICATE

This is to certify that the report titled **Performance Prediction and Optimization of Supercapacitors Through Data-Driven Analysis**, submitted by **Ashish Pratap Singh (EC21B1092)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bona fide record of the work done by him/her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Supercapacitors have emerged as a leading class of energy storage devices, offering high power density, rapid charge-discharge capabilities, and long cycle life. However, optimizing their performance through conventional experimentation is time-consuming and resource-intensive due to the vast combinatorial space of materials and testing conditions. This project explores a machine learning-based approach to enhance supercapacitor design by addressing two key challenges: (1) predicting areal capacitance based on device configuration and operating parameters, and (2) recommending optimal combinations of substrate, electrodes, electrolytes, and electrolyte concentration to achieve desired performance metrics such as capacitance, energy density, and power density.

The first part of the study employed regression algorithms including Support Vector Regression (SVR), Decision Tree, XG Boost, Linear Regression and Random Forest Regression to estimate areal capacitance using input features such as specific surface area, pore width, pore volume, pore size, Id/Ig ratio, N% and O%. The second part utilized models such as Decision Trees, Random Forests, XG Boost and Gradient Boosting to identify ideal material combinations for target performance ranges. Results from the capacitance prediction models showed excellent agreement between predicted and actual values, with R^2 scores of 0.95 for the best model. Material prediction models achieved accuracy exceeding 99% in predicting suitable material combinations. This dual-approach methodology highlights the feasibility of using machine learning for performance optimization and accelerated materials discovery in supercapacitor research.

KEYWORDS: Supercapacitor; Carbon based electrode; Machine Learning.

TABLE OF CONTENTS

ACKNOWLEDGEMENTS **i**

ABSTRACT **ii**

LIST OF TABLES **v**

LIST OF FIGURES **vi**

ABBREVIATIONS **vii**

1 Introduction **1**

1.1 Introduction 1

1.1.1 Background 1

1.1.2 Motivation and Problem Statement 1

1.1.3 Objectives of the Study 2

1.1.4 Scope of the Report 3

1.1.5 Organization of the Report 3

1.2 Chapter Summary 4

2 Literature Review **5**

2.1 Supercapacitor Fundamentals 5

2.1.1 Types of Supercapacitors 5

2.1.2 Key Performance Metrics 6

2.1.3 Applications of Supercapacitors 7

2.2 Influence of Electrodes and Electrolytes 7

2.2.1 Electrode Materials 8

2.2.2 Electrolyte Selection 8

2.2.3 Effect of Electrode-Electrolyte Interactions 8

2.3 Applications of ML in Material Science 9

2.4 Chapter Summary 9

3 METHODOLOGY **10**

3.1 Data Collection and Preprocessing 10

3.1.1 Sources of Data 10

3.1.2 Capacitance Prediction Dataset 10

3.1.3 Material Prediction Dataset 11

3.2 Data Analysis 12

3.2.1 Capacitance Prediction Data Analysis 12

3.2.2 Material Prediction Data Analysis 16

3.3 ML Algorithms 20

3.3.1 Machine Learning Models for Capacitance Prediction . . . 20

3.3.2 Machine Learning Models for Material Prediction 21

3.3.3 Evaluation Metrics 22

3.4 Chapter Summary 22

4 RESULTS AND DISCUSSION **23**

4.1 Capacitance Prediction 23

4.1.1 Performance Comparison 23

4.2 Material Prediction 26

4.2.1 Performance Comparison 26

4.3 Chapter Summary 29

5 CONCLUSION AND FUTURE SCOPE **30**

5.1 Conclusion 30

5.2 Future Scope 31

5.3 Chapter Summary 32

LIST OF TABLES

3.1 Capacitance Prediction Dataset [1]. 11

3.2 Dataset used for Material Prediction - Part 1 12

3.3 Dataset used for Material Prediction - Part 2 13

3.4 Characteristics of Dataset used for Capacitance Prediction 14

3.5 Characteristics of Dataset used for Material Prediction 17

4.1 Performance comparison of different ML models 24

4.2 Comparison of different regression models 28

LIST OF FIGURES

2.1 Comparison of Electrochemical Double-Layer Capacitors (EDLCs)(a), Hybrid Supercapacitors (b), and Pseudocapacitors (c) [2]. 6

3.1 Scatter plots showing the relationship between Capacitance (F/g) and four input features: PW (V), SSA, PV, and PS. 14

3.2 Scatter plots illustrating the relationship between Capacitance (F/g) and two chemical structure features: ID/IG and nitrogen content (N%). 15

3.3 Scatter plot showing the relationship between Capacitance (F/g) and oxygen content (O%). 16

3.4 Distribution of different types of electrode (left) and electrolyte (right) materials. 18

3.5 Distribution of different types of substrate materials. 18

3.6 Pearson correlation heatmap showing the relationship among key electrochemical features. Strong positive and negative correlations are represented by red and blue colors respectively, with the intensity indicating the strength of correlation. 19

4.1 Actual vs. Predicted Capacitance for various models: XGBoost, Random Forest, Decision Tree, Neural Net (MLP), SVR (Tuned), Linear Regression (Polynomial), and the final Stacked Model. 25

4.2 Demonstrating the user interface and interaction flow. 27

ABBREVIATIONS

EDLC	Electric Double Layer Capacitance
ML	Machine Learning
SSA	Specific Surface Area
PV	Pore Volume
PW	Potential Window
PS	Pore Size
ID/IG	Ratio of D-band to G-band in Raman Spectroscopy (Indicates Structural Defects)
PEEK	polyetheretherketone

CHAPTER 1

Introduction

1.1 Introduction

1.1.1 Background

Supercapacitors have emerged as promising energy storage solutions because of their exceptional power density, fast charging and discharging rates, and extended operational lifespan. Unlike traditional batteries, which store energy through chemical reactions, supercapacitors rely on electrostatic charge separation, allowing them to deliver quick bursts of energy [3]. This makes them particularly useful in applications where rapid energy transfer is needed, such as electric vehicles, renewable energy storage, and portable electronics. With growing demands, optimizing supercapacitor performance has become a crucial area of research [4].

The performance of supercapacitors is heavily influenced by the choice of electrode material, electrolyte composition, and electrolyte concentration. Researchers have explored various materials, including carbon-based structures (graphene, carbon nanotubes), transition metal oxides, and conductive polymers, to improve capacitance and energy density [3]. Similarly, electrolytes play a key role in determining ionic conductivity and electrochemical stability, further impacting the overall efficiency of the device [5].

1.1.2 Motivation and Problem Statement

The demand for lightweight, flexible, and high-performance energy storage devices is growing rapidly with the expansion of wearable electronics, portable sensors, and flexible IoT systems. Among various energy storage technologies, supercapacitors have emerged as a promising candidate due to their high power density, long cycle life, and fast charge/discharge characteristics. Despite these advantages, improving energy density and optimizing the performance of supercapacitors under flexible configurations remains a major challenge. Traditional methods of supercapacitor design involve trial-and-error experimentation, which can be costly and time-consuming, especially given the vast space of possible electrode, electrolyte, and substrate combinations. In this context, machine learning (ML) has emerged as a powerful tool for predicting device performance and identifying optimal material combinations without the need for exhaustive experimentation. By training ML models on existing experimental data, it becomes possible to accelerate the design of next-generation supercapacitors through data-driven insights. This project leverages machine learning techniques to address two key challenges in supercapacitor research: predicting the areal capacitance based on material and experimental parameters, and identifying the most promising combinations of electrode, electrolyte, electrolyte concentration and substrate materials to meet specific performance targets.

The development of efficient and flexible supercapacitors is hindered by the complexity of selecting suitable materials and operating conditions that yield desirable electrochemical performance. The lack of comprehensive databases and the experimental cost of testing every possible material configuration make the optimization process slow and inefficient. This project addresses the following two central problems:

1. **Capacitance Prediction:** How can we accurately predict the areal capacitance of a supercapacitor device based on its configuration?
2. **Material Combination Recommendation:** Given a desired performance range (e.g., energy density, power density, areal capacitance), which combination of electrode, electrolyte, electrolyte concentration and substrate materials is most likely to meet the requirement?

Solving these problems using machine learning offers a pathway to accelerate supercapacitor research and reduce dependence on physical prototyping.

1.1.3 Objectives of the Study

The objectives of this project are twofold, focusing on the application of machine learning techniques to enhance the design and performance optimization of supercapacitor devices:

1. **Capacitance Prediction:** The first objective is to develop robust machine learning regression models capable of accurately predicting the areal capacitance of supercapacitors. By learning from patterns in experimental data, the models aim to facilitate rapid estimation of areal capacitance without the need for extensive physical prototyping or testing.

2. **Material Recommendation:** The second objective is to construct machine learning models that can intelligently recommend optimal combinations of electrode, electrolyte, electrolyte concentration and substrate materials to achieve specific target ranges for crucial performance metrics. These include energy density, power density, and areal capacitance three vital indicators of supercapacitor efficiency and long-term reliability. The regression framework enables guided material selection based on desired performance outcomes, aiding researchers in narrowing down experimental efforts.

1.1.4 Scope of the Report

This report presents a comprehensive methodology for applying machine learning to supercapacitor research. It includes a literature review covering supercapacitor fundamentals, material selection, and the role of machine learning in material science. The study details the complete process of data collection, preprocessing, and machine learning model development. Several machine learning models have been successfully implemented—both for predicting capacitance based on material and process parameters, and for identifying optimal material combinations to achieve target values of areal capacitance, energy density, and power density. The report further analyzes the results obtained from these models and explores future directions for enhancing material discovery through machine learning. The findings provide a solid foundation for advancing predictive tools aimed at optimizing supercapacitor performance.

1.1.5 Organization of the Report

This report is organized into five chapters, each addressing a key component of the study. The structure is as follows:

Chapter 1: Introduction

This chapter introduces the motivation for the project, outlining the growing importance of supercapacitors in flexible energy storage applications. It defines the problem statement, objectives, and scope of the work, highlighting the role of machine learning in optimizing supercapacitor performance.

Chapter 2: Literature Review

A comprehensive review of the literature is provided, covering the fundamentals of supercapacitors, types of electrode and electrolyte materials, key performance metrics, and recent advances in the application of machine learning in materials science.

Chapter 3: Methodology

This chapter details the data collection and preprocessing techniques used to construct two datasets—one for capacitance prediction and the other for material recommendation. It also explains the machine learning models implemented, feature analysis, and the evaluation metrics employed.

Chapter 4: Results and Discussion

The performance of various machine learning models is analyzed and discussed in this chapter. Results for both capacitance prediction and material recommendation tasks are presented, along with comparative analyses and key observations based on the evaluation metrics.

Chapter 5: Conclusion and Future Scope

The final chapter summarizes the outcomes of the study, emphasizes the effectiveness of the selected models, and outlines future directions, including dataset expansion and real-world validation of model predictions.

1.2 Chapter Summary

This chapter introduces the need for high-performance, flexible energy storage systems and the challenges in optimizing supercapacitor configurations using traditional trial-and-error methods. It highlights the role of machine learning as a predictive and prescriptive tool to enhance design efficiency. The key problems addressed include predicting areal capacitance and recommending optimal material combinations.

CHAPTER 2

Literature Review

2.1 Supercapacitor Fundamentals

Supercapacitors, also known as electrochemical capacitors, are energy storage devices that bridge the performance gap between conventional capacitors and rechargeable batteries. Instead of depending on chemical transformations like batteries, they store energy through the accumulation of electrostatic charge. This mechanism allows for rapid charging and discharging, high power delivery, and long operational life[3].

2.1.1 Types of Supercapacitors

Based on how they store charge, supercapacitors are generally divided into three categories. One type, known as Electrochemical Double-Layer Capacitors (EDLCs), stores energy through the physical separation of charges at the boundary between the electrode surface and the electrolyte. Materials like activated carbon and graphene are widely utilized due to their extensive surface area and superior electrical conductivity [3]. Pseudocapacitors, in contrast, store energy through faradaic charge transfer involving redox reactions. Transition metal oxides such as MnO_2 and RuO_2, along with conductive polymers, are commonly used as electrode materials due to their excellent electrochemical properties [6]. Lastly, Hybrid Supercapacitors combine the strengths of EDLCs and pseudocapacitors by integrating both electrostatic and faradaic charge storage mechanisms. An example of this is asymmetric supercapacitors, where one electrode functions as an EDLC and the other as a pseudocapacitor [7]. Figure 2.1 illustrates all three types of supercapacitors.

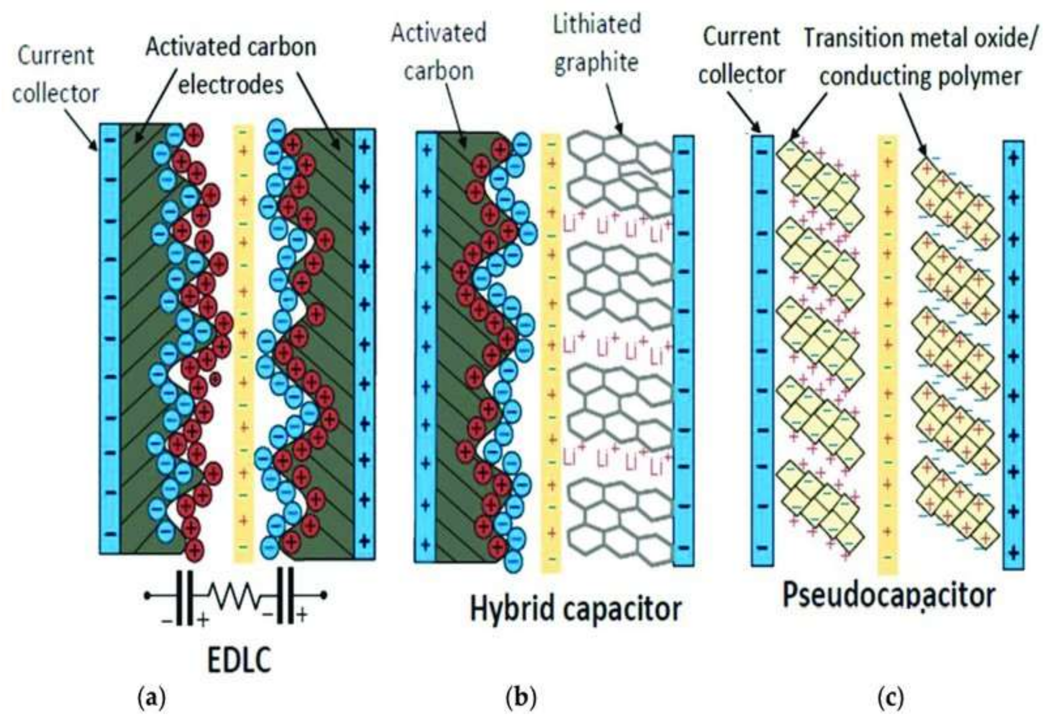


Figure 2.1: Comparison of Electrochemical Double-Layer Capacitors (EDLCs)(a), Hybrid Supercapacitors (b), and Pseudocapacitors (c) [2].

2.1.2 Key Performance Metrics

Several parameters define the efficiency and performance of supercapacitors:

- **Capacitance (C):** The ability of a supercapacitor to store charge, measured in farads (F). It depends on the surface area and porosity of electrode materials.
- **Energy Density (E):** Indicates the amount of energy stored per unit volume or mass, given by the equation:

$$E = \frac{1}{2}CV^2 \tag{2.1}$$

  where is the operating voltage.
- **Power Density (P):** Determines how quickly the stored energy can be delivered, crucial for applications requiring high bursts of power.
- **Cyclability:** Indicates lifespan of a supercapacitor, measured by the number of charge-discharge cycles it can endure before experiencing significant performance deterioration. [3].

2.1.3 Applications of Supercapacitors

Due to their fast charge-discharge capability, supercapacitors are widely used across various applications. In the context of renewable energy systems, supercapacitors contribute to grid stability by capturing surplus energy during periods of high solar or wind output and supplying it when generation drops. This boosts overall efficiency and helps lower reliance on fossil fuels. In electric vehicles (EVs), they support rapid acceleration and recover energy during braking, working alongside lithium-ion batteries to manage sudden power surges and enhance battery lifespan[8]. They also play a crucial role in consumer electronics, serving as power backups for memory chips, smartwatches, and portable medical devices, ensuring uninterrupted operation due to their rapid charging ability and sustained power delivery [9]. Additionally, in public transportation systems, supercapacitors are increasingly integrated into buses and trams, enabling quick charging at stops [10] and reducing reliance on fuel-powered transportation. In industrial power management, they are used in cranes, forklifts, and robotic systems to smooth power fluctuations and improve efficiency, ensuring stable and reliable operation in high-power-demand environments [4].

2.2 Influence of Electrodes and Electrolytes

The performance characteristics of a supercapacitor such as its capacitance, energy density, and overall storage efficiency are heavily influenced by the selection of electrode materials and electrolytes. Research has shown that electrodes made from MXenes offer remarkable electrical conductivity along with strong electrochemical stability [11]. Extensive research has focused on materials like graphene oxide and transition metal oxides, including MnO_2 and RuO_2, due to their exceptional charge storage performance [12]. While electrode materials largely determine how electrical charge is accumulated, electrolytes are essential for facilitating ion movement and preserving the system's stability over time.

2.2.1 Electrode Materials

Carbon-Based Electrodes such as activated carbon, graphene, and carbon nanotubes (CNTs) are widely used due to their high surface area, tunable porosity, and excellent electrical conductivity. ese materials are mainly used in electrochemical double-layer capacitors (EDLCs), where energy storage occurs via the separation of charges instead of relying on chemical reactions [3]. Metal Oxides, including MnO, RuO, and $NiCoS$, provide high capacitance due to their redox reactions, making them ideal for pseudo-capacitive applications. Among these, RuO exhibits one of the highest capacitances but is expensive, leading to research on cost-effective alternatives [12]. A novel family of two-dimensional transition metal carbides and nitrides, referred to as MXenes, exhibit metallic conductivity and hydrophilic surfaces, enhancing ion diffusion and charge storage efficiency, showing promising results in hybrid supercapacitors [11].

2.2.2 Electrolyte Selection

Electrolytes are essential for defining the voltage range, ion transport efficiency, and overall stability of a supercapacitor. They can be categorized into various types depending on their composition and performance attributes. Aqueous electrolytes are typically composed of acids (HSO), bases (KOH), or neutral salts ($NaSO$). These electrolytes offer high ionic conductivity but have a limited voltage window of approximately 1V due to water decomposition [13]. Organic electrolytes, based on acetonitrile or propylene carbonate solvents, provide a higher voltage window (2.7–3V), enabling higher energy densities. However, they exhibit lower ionic conductivity and pose flammability concerns [14]. Ionic liquids are non-volatile, thermally stable electrolytes that enable ultrahigh voltage operation (>4V). Their high viscosity reduces ionic mobility, but ongoing research is improving their performance [15].

2.2.3 Effect of Electrode-Electrolyte Interactions

The interplay between electrode materials and electrolytes directly affects the charge storage mechanism and efficiency. Research has indicated that for optimal charge storage, the pore size distribution of carbon electrodes must align with the ionic size of the electrolyte [16]. MXene-based electrodes exhibit excellent compatibility with aqueous and organic electrolytes due to their hydrophilic nature [13]. Pseudocapacitive materials like MnO perform best in alkaline electrolytes (e.g., KOH) due to enhanced redox reaction kinetics. Despite their low conductivity, ionic liquids are ideal for high-voltage applications due to their wide electrochemical stability window [15].

2.3 Applications of ML in Material Science

Machine Learning (ML) is making a big impact in several areas of material science. In the past, scientists had to rely on trial-and-error experiments, which were expensive and time-consuming. Now, with ML, computers can quickly analyze data and find patterns that help predict material properties, reducing the need for excessive lab work. It is widely used in predicting material properties, where algorithms can estimate characteristics such as conductivity, stability, and capacitance by learning from previous experiments [17]. ML also plays a crucial role in finding new materials by exploring massive databases and suggesting materials with desired properties, accelerating the discovery process for researchers [18]. Additionally, ML aids in improving manufacturing processes by adjusting variables like temperature and pressure in real-time to optimize material performance [19]. Another important application is detecting defects, where computer vision powered by ML can identify defects in materials during production, enhancing quality control and reducing waste [20].

2.4 Chapter Summary

This chapter provides a foundational understanding of supercapacitor technologies, including types, working mechanisms, and performance metrics such as capacitance, energy density, and power density. It also surveys various electrode and electrolyte materials and discusses the increasing relevance of machine learning in materials research, particularly in performance prediction and optimization.

CHAPTER 3

METHODOLOGY

3.1 Data Collection and Preprocessing

The performance and precision of a machine learning model are heavily influenced by the dataset utilized for both training and testing. For this project, data has been collected from multiple research papers and experimental studies that report the capacitance values of different electrode and electrolyte combinations, along with other material properties.

3.1.1 Sources of Data

The data for this project has been gathered from:

- **Published Research Papers:** Journals and conference proceedings that provide detailed experimental results of different supercapacitor materials.
- **Supplementary Materials from Scientific Articles:** Some articles include additional datasets used in their studies, providing valuable numerical data.
- **Gaussian Noise Technique:** To address the limitation of a small dataset, the Gaussian noise technique described in [21] was applied to generate an extended dataset.

3.1.2 Capacitance Prediction Dataset

Table 3.1 presents a subset of the collected data used for capacitance prediction. Initially sourced from the research paper by Su et al. (2019) [1], the dataset contained 121 rows, which were later expanded to 1200 rows using the Gaussian noise technique described in the research paper by Bilali et al. (2021)[21].

Table 3.1: Capacitance Prediction Dataset [1].

PW (V)	SSA (m^2/mg)	PV (cm^3/g)	PS (nm)	ID/IG	N% (at.%)	O% (at.%)	Capacitance (F/g)
0.8	0.68	0.72	4	1.62	0	10.75	100
0.8	1.38	1.2	1	1.3	0	9	160
1	0.38	0.51	5.4	1.15	1.53	3.1	227
1	2.01	1.6	0.55	1.12	0.1	6.27	319
1	0.34	1.6	0.62	1.38	0	1.54	7.86
1	0.3	1.5	0.62	1.25	0	3.67	19.9
1	0.34	1.59	0.62	1.22	0.54	1.8	22.2
1	0.34	1.34	0.62	1.03	1.02	2.85	29.7
1	0.34	1.6	0.62	1.38	0	1.54	2.59
1	0.3	1.5	0.62	1.25	0	3.67	34.6

3.1.3 Material Prediction Dataset

The dataset includes various properties that influence capacitance and energy storage efficiency, which are essential for predicting the optimal electrode, electrolyte, electrolyte concentration, and substrate required to achieve a specific target value. The choice of electrode material is critical in influencing capacitance, power density, and energy density. Materials like graphene, RuO_2, MXene, and carbon nanotubes (CNTs) are widely utilized for their exceptional surface area and electrical conductivity [22]. Similarly, the electrolyte type impacts the ionic conductivity and operating voltage range of a supercapacitor, with commonly used electrolytes including aqueous solutions such as H_2SO_4 and KOH, as well as organic solvents and ionic liquids. Polyvinyl alcohol (PVA) serves as a versatile matrix for gel electrolytes due to its excellent film-forming ability and mechanical stability. When combined with acids like sulfuric acid (H_2SO_4) or phosphoric acid (H_3PO_4), PVA-based GPEs exhibit high ionic conductivity and flexibility. For instance, PVA-H_2SO_4 gel electrolytes have demonstrated promising ionic conductivity and specific capacitance performance, making them suitable for flexible energy storage devices [23]. The electrolyte concentration (g/L) directly affects charge transport and ion mobility, influencing both capacitance and overall energy storage efficiency.

Key performance metrics include capacitance (mF/cm^2), which quantifies a supercapacitor's ability to store charge, and energy density (μWh/cm^2), representing the total energy stored per unit area, which depends on capacitance and operating voltage. Power density (mW/cm^2) is another key factor, as it dictates the rate at which energy can be delivered. A higher power density allows for faster charge-discharge cycles, making the device well-suited for high-power applications. Furthermore, cyclability (%) indicates how well the supercapacitor retains its performance after multiple charge-discharge cycles, with a higher percentage of cyclability reflecting better long-term stability. The number of cycles determines the lifespan of a supercapacitor, referring to the total number of charge-discharge cycles it can undergo before experiencing significant degradation. Tables 3.2 and 3.3 present a subset of the collected data used for the analysis.

Table 3.2: Dataset used for Material Prediction - Part 1

S. No.	Electrode	Substrate	Electrolyte	Electrolyte Conc. (g/l)	Capacitance (mF/cm^2)	Current (mA/cm^2)
1	LIG	Kapton PI	PVA-H3PO4	85	19.47	0.04
2	2% CNT	Kapton PI	PVA/H2SO4	125	51.975	0.2
3	5% CNTs	Kapton PI	PVA/H2SO4	125	28.375	0.2
4	GO-MnO2	Kapton PI	PVA-H3PO4	5	4.02	0.5
5	LIG-MXene	Kapton PI	PVA-H3PO4	-	2.19	0.1
6	LIG-C	Parylene-C	PVA/H2SO4	85.87	1.66	0.5
7	B-doped LIG	Kapton PI	PVA/H2SO4	163.49	16.5	0.05
8	RuO2/Au	Kapton PI	PVA/H2SO4	90.9	27	2.7
9	NiCo2S4	Kapton PI	Co-Ni-Thiourea	228.36	30.4	0.75
10	d-LIG-D	Kapton PI	PVA/H2SO4	98.08	19.5	0.05
11	LIG	Whatman Paper	PVA/H2SO4	87.62	4.6	0.5
12	LIG	Lignin	PVA/H2SO4	98.08	0.88	0.02
13	LIG	Cork	PVA/H2SO4	87.62	1.35	0.05

3.2 Data Analysis

3.2.1 Capacitance Prediction Data Analysis

The dataset comprises 1200 records, each characterized by eight distinct features that vary in scale and complexity. This moderate size is ideal for training a wide range of machine learning models, as it balances computational efficiency with sufficient variability to uncover meaningful patterns. Among the features, ’PS’ and ’Capacitance’ demonstrate high variability, with means of 3.15 and 250.6, and standard deviations of 4.81 and

Table 3.3: Dataset used for Material Prediction - Part 2

Scan Rate (mV/s)	Energy Density (μWh/cm^2)	Power Density (mW/cm^2)	Cyclability (%)	No. of Cycles	Reference
2	3.38	0.199	82	1000	[3]
2	6.5	0.219	-	-	[24]
2	5.01	0.199	-	-	[24]
10	1286.4	-	92.59	1000	[12]
10	-	-	102.4	1000	[11]
10	0.19	0.00459	96	10000	[25]
10	-	-	90	12000	[26]
100	3.1	2.3	80	10000	[27]
5	16.9	0.368	84	1000	[28]
2	2.7	0.025	87.4	10000	[29]
10	0.3	0.0045	96	10000	[30]
10	0.152	0.025	91	10000	[31]
5	0.2	0.08	106	10000	[32]

118.53, respectively. Such elevated levels of dispersion suggest that these features may exhibit nonlinear relationships or interactions that require flexible modeling techniques. Conversely, features like ’PW’ (mean: 0.98, SD: 0.19) and ’ID/IG’ (mean: 1.09, SD: 0.44) show more constrained distributions, indicating potentially more straightforward or stable contributions to the target variable. The dataset also includes compositional attributes such as ’N%’ and ’O%’, which show moderate to high variance (SDs of 3.83 and 5.34, respectively), pointing to differences in material chemistry that could significantly affect electrochemical performance. The presence of noise in the dataset can be inferred from the standard deviation values, with features like ’Capacitance’ showing substantial measurement fluctuation, while ’PW’ remains relatively stable. Such variation underscores the importance of using machine learning models capable of handling heterogeneous data patterns and mitigating the influence of noise—particularly through techniques such as regularization. Overall, the diversity and spread of the features reflect a complex underlying structure, which can be effectively explored through robust modeling approaches. The detailed summary of these characteristics is provided in Table 3.4.

Figure 3.1 illustrates scatter plots depicting the relationship between the target variable, capacitance (F/g), and four selected input features: PW (V), SSA, PV, and PS. The

Table 3.4: Characteristics of Dataset used for Capacitance Prediction

	Feature Mean	Std Dev	Min	25 %	50 %	75 %	Max	Noise level (Std Dev)
PW (V)	0.98	0.19	-0.37	0.9	1.0	1.05	1.87	0.19
SSA	1.32	0.94	-0.3	0.47	1.22	1.83	4.78	0.94
PV	1.09	0.76	-0.2	0.62	0.96	1.5	5.9	0.76
PS	3.15	4.81	0.03	1.27	2.17	2.99	36.25	4.81
ID/IG	1.09	0.44	-0.59	0.85	1.01	1.24	4.01	0.44
N%	3.28	3.83	-0.54	0.6	1.6	5.36	20.63	3.83
O%	8.69	5.34	-0.25	4.6	7.48	10.99	25.3	5.34
Capacitance (F/g)	250.6	118.53	-14.28	185.08	262.22	313.53	567.26	118.53

distribution of data points in the Capacitance vs. PW (V) plot shows that capacitance values are moderately clustered within a specific voltage range, suggesting a potential but not strongly linear relationship. In the Capacitance vs. SSA plot, the points are widely dispersed, with capacitance values appearing throughout the range of SSA, which indicates that surface area alone may not be a strong independent predictor of capacitance. A similar pattern is seen in the Capacitance vs. PV plot, where although

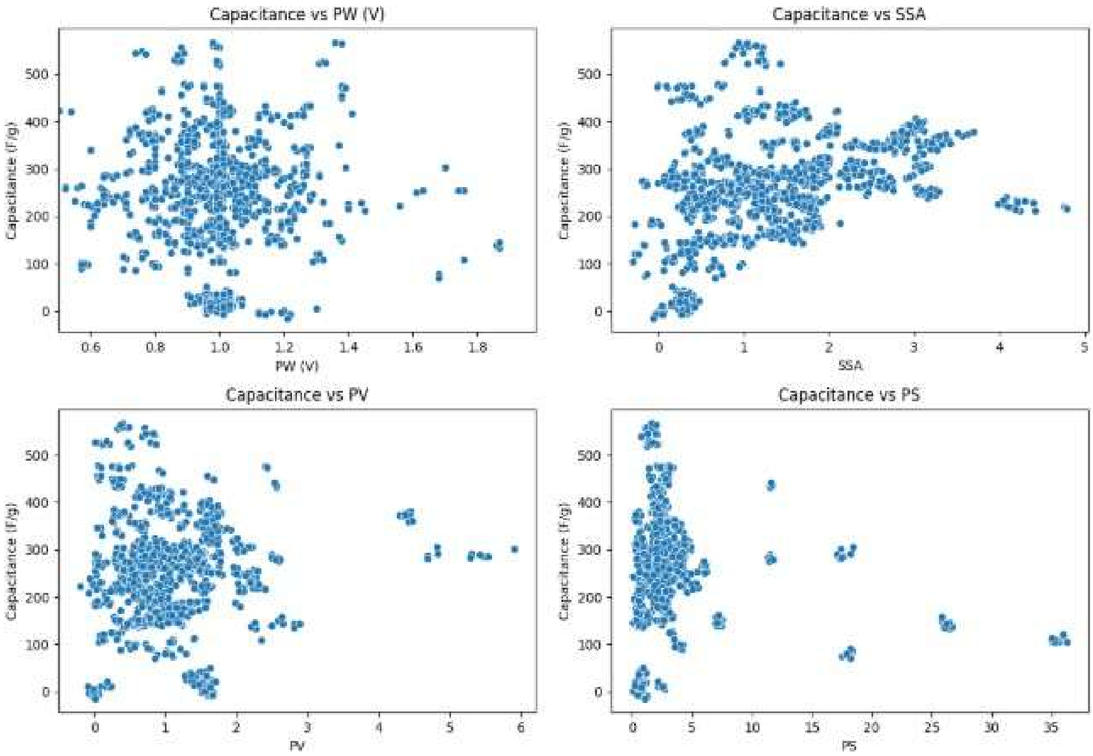


Figure 3.1: Scatter plots showing the relationship between Capacitance (F/g) and four input features: PW (V), SSA, PV, and PS.

a large number of points concentrate at lower pore volume values, high capacitance is achieved across varying levels of PV, implying nonlinear behavior or the influence of other interacting variables. Notably, the Capacitance vs. PS plot exhibits a dense vertical cluster of data at lower pore size values, with only a few data points scattered at higher pore sizes. This could indicate that most materials in the dataset have relatively small pores and that larger pore sizes do not consistently correspond to higher capacitance. Overall, the visualizations reveal a complex, non-linear relationship between capacitance and these physical characteristics, reinforcing the need for machine learning models capable of capturing such patterns beyond traditional linear assumptions.

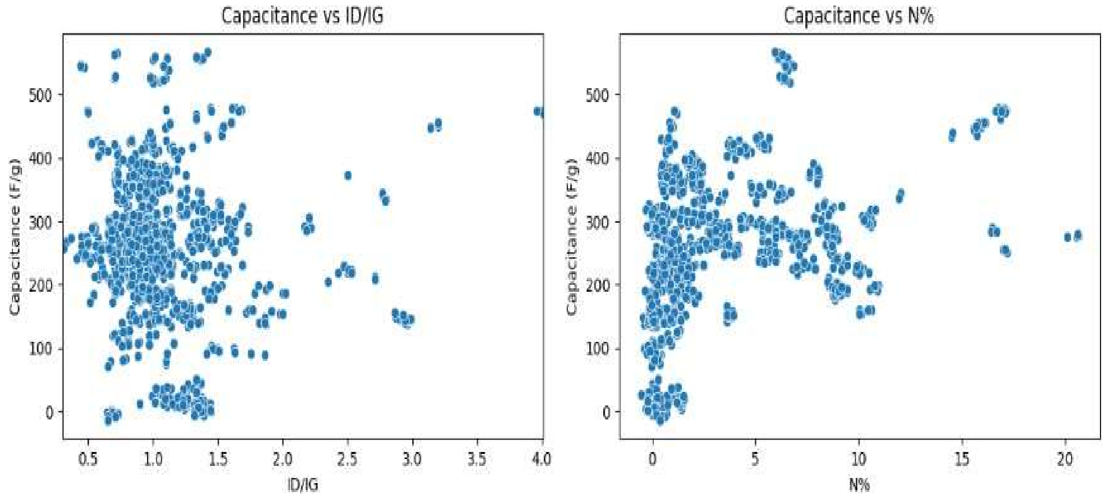


Figure 3.2: Scatter plots illustrating the relationship between Capacitance (F/g) and two chemical structure features: ID/IG and nitrogen content (N%).

Figure 3.2 presents scatter plots that explore the relationship between capacitance and two structural or compositional features: the Raman intensity ratio (ID/IG) and nitrogen content (N%). In the Capacitance vs. ID/IG plot, the data points are largely concentrated between ID/IG values of 0.9 and 1.5, suggesting that most materials fall within this range. While capacitance values are scattered broadly across the vertical axis, some clustering around mid-range capacitance levels can be observed. The lack of a clear trend implies that the ID/IG ratio, which reflects the degree of disorder in carbon materials, may influence capacitance but not in a strictly linear fashion. Similarly, in the Capacitance vs. N% plot, capacitance shows a wide distribution over varying nitrogen concentrations. Although higher nitrogen content appears in fewer samples, it is associated with both low and high capacitance values, indicating that nitrogen doping might contribute positively to performance, but its effect may depend on interactions with other factors such as pore structure or surface area. The dispersion observed in both plots reinforces the idea that these features impact capacitance in a complex, possibly nonlinear way, and highlights the relevance of machine learning techniques that can model such interactions more effectively than traditional methods.

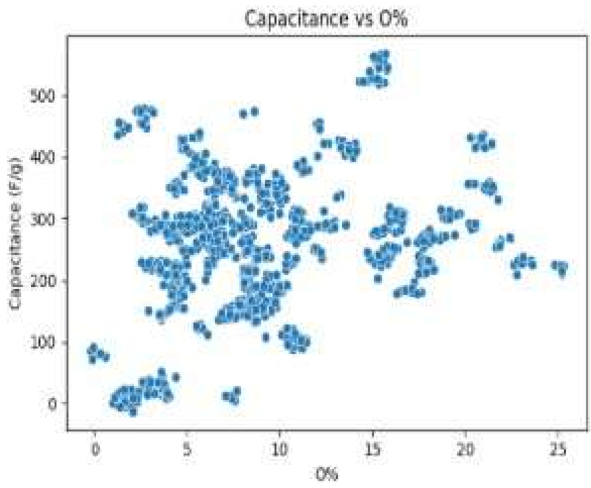


Figure 3.3: Scatter plot showing the relationship between Capacitance (F/g) and oxygen content (O%).

Figure 3.3 presents a scatter plot depicting the relationship between capacitance (F/g) and oxygen content (O%) in the materials. The data points are broadly distributed across the entire range of oxygen concentrations, from near 0% up to approximately 25%. A noticeable concentration of samples appears between 5% and 15% oxygen content, but capacitance values remain widely spread across this interval. While no clear linear trend is evident, the plot suggests that moderate levels of oxygen content are frequently associated with both average and high capacitance values, indicating that oxygen functional groups may play a role in enhancing electrochemical performance. However, the presence of high capacitance values even at lower or higher oxygen percentages implies that this feature does not solely govern capacitance outcomes. Instead, its influence might be context-dependent, interacting with other structural or compositional factors. The overall dispersion of data reinforces the complexity of the relationship, underscoring the need for modeling approaches capable of identifying nonlinear patterns and feature interactions within the dataset.

3.2.2 Material Prediction Data Analysis

Table 3.5 provides a summary of the statistical characteristics of the dataset used in this study, including the mean, standard deviation, minimum, 25th percentile, median (50th percentile), 75th percentile, maximum values, and noise level (standard deviation) for each feature. The electrolyte concentration has a mean of 177.536 g/l, with a standard deviation of 94.296, indicating significant variability across the dataset, ranging from 76.475 g/l to 345.444 g/l. The areal capacitance exhibits considerable variability, with a mean of 24.614 mF/cm² and a standard deviation of 25.255, spanning from 0.005 mF/cm² to 101.930 mF/cm². The current has a mean of 0.526 mA/cm² and a standard deviation of 0.716, with values ranging from a minimum of 0.003 mA/cm² to a maximum of 2.808 mA/cm². The scan rate shows a wide spread, with a mean of 17.789 mV/s and a standard deviation of 27.951, ranging from 0.100 mV/s to 103.333 mV/s. The energy density has a mean of 4.048 uWh/cm² and a standard deviation of 4.818, with values ranging from 0.003 uWh/cm² to 17.476 uWh/cm². Similarly, power density shows variability, with a mean of 0.616 mW/cm² and a standard deviation of 0.856, ranging from 0.002 mW/cm² to 2.591 mW/cm². The cyclability feature, with a mean of 94.649%, exhibits a high degree of variation, with a standard deviation of 23.592%, and values spanning from 73.047% to 192.398%. Finally, the number of cycles has a mean of 7400.199 cycles, with a standard deviation of 3834.446, reflecting a wide range of cycle numbers from 623.018 to 10443.679 cycles. The noise level, indicated by the standard deviation for each feature, illustrates the inherent variability present in the dataset, which must be taken into account during analysis and model predictions.

Table 3.5: Characteristics of Dataset used for Material Prediction

Feature	**Mean**	**Std Dev**	**Min**	**25 %**	**50 %**	**75 %**	**Max**	**Noise level (Std Dev)**
Electrolyte Conc. (g/l)	177.536	94.296	76.475	89.136	183.844	276.440	345.444	94.296
Areal Capacitance (mF/cm²)	24.614	25.255	0.005	4.350	20.344	28.657	101.930	25.255
Current (mA/cm²)	0.526	0.716	0.003	0.068	0.463	0.526	2.808	0.716
Scan Rate (mV/s)	17.789	27.951	0.100	3.898	8.292	11.228	103.333	27.951
Energy Density (µWh/cm²)	4.048	4.818	0.003	0.468	2.768	3.595	17.476	4.818
Power Density (mW/cm²)	0.616	0.856	0.002	0.074	0.272	0.616	2.591	0.856
Cyclability (%)	94.649	23.592	73.047	81.736	92.821	97.028	192.398	23.592
No. of cycles	7400.199	3834.446	623.018	2816.714	9864.716	10031.712	10443.679	3834.446

The left plot of Figure 3.4 reveals that Laser-Induced Graphene (LIG) is the most frequently utilized electrode material, significantly surpassing all others in usage. This trend underscores LIG's prominence in current research, likely due to its excellent conductivity, flexibility, and cost-effective fabrication. LIG/MnO_2 and RuO_2/Au appear

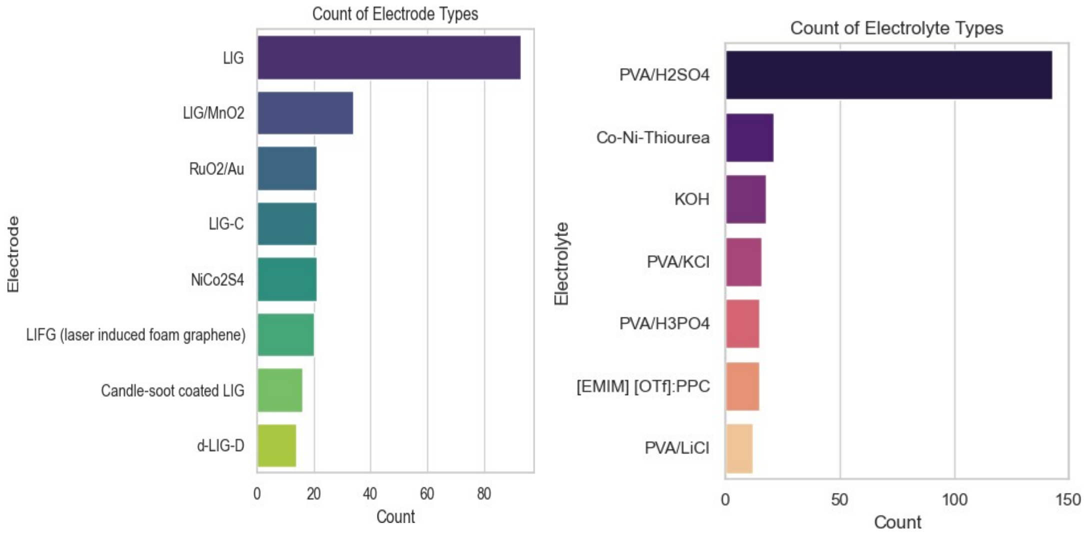


Figure 3.4: Distribution of different types of electrode (left) and electrolyte (right) materials.

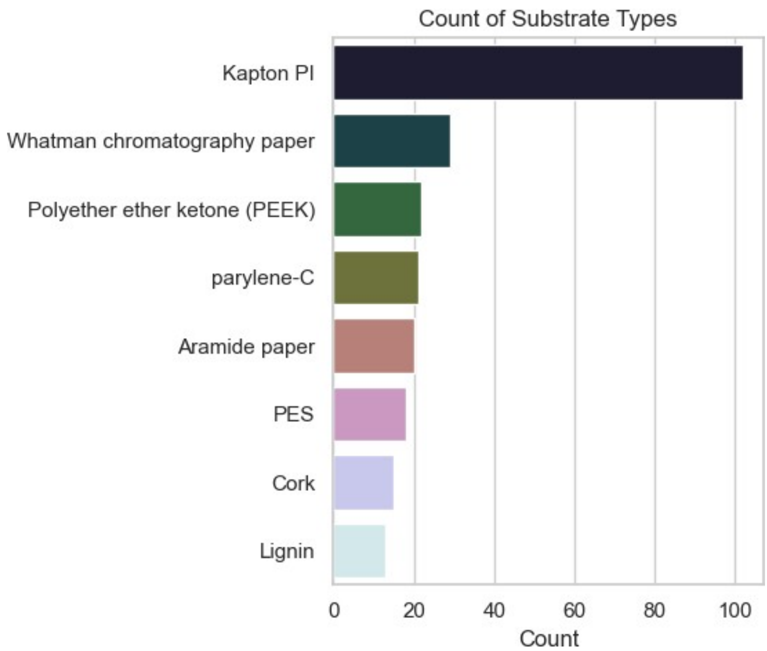


Figure 3.5: Distribution of different types of substrate materials.

next in popularity, reflecting the interest in hybrid structures that enhance electrochemical performance through pseudocapacitive contributions. The right plot of Figure 3.4 shows the dominance of PVA/H_2SO_4 as the electrolyte. Its high frequency suggests that this polymer gel electrolyte is favored for its stability, ionic conductivity, and compatibility with flexible energy storage devices. In the Figure 3.5, Kapton PI stands out as the most common substrate, likely due to its thermal stability, chemical resistance, and excellent mechanical properties. Other substrates such as Whatman chromatography paper, PEEK, and parylene-C are also represented, pointing to a diverse range of materials being explored for flexible and printable supercapacitor configurations. The wide distribution of substrate types reflects the ongoing experimentation to optimize device flexibility, durability, and compatibility with various fabrication techniques. Together, these plots offer insight into current material preferences in supercapacitor research and provide a foundation for understanding how these selections may influence device performance.

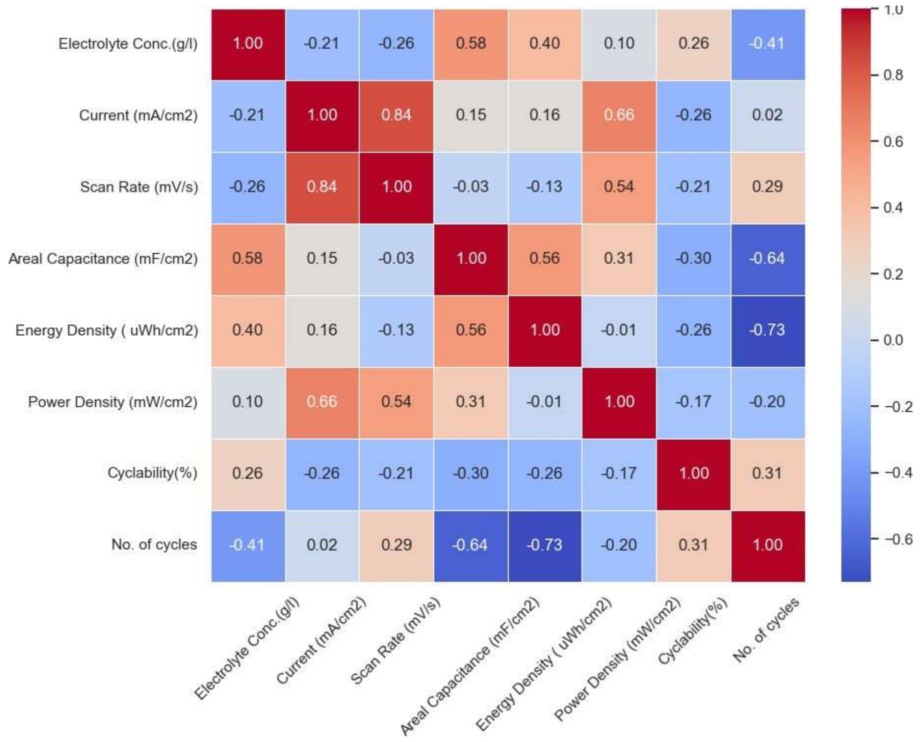


Figure 3.6: Pearson correlation heatmap showing the relationship among key electrochemical features. Strong positive and negative correlations are represented by red and blue colors respectively, with the intensity indicating the strength of correlation.

The correlation heatmap shown in Figure 3.6 provides key insights into the interdependencies between various parameters affecting supercapacitor performance. Notably, current and scan rate exhibit a strong positive correlation (0.85), highlighting their direct relationship during electrochemical measurements. Areal capacitance is positively influenced by both electrolyte concentration (0.60) and energy density (0.61), confirming their significance in energy storage optimization. Conversely, a strong negative correlation between the number of cycles and energy density (-0.75) indicates a trade-off between performance and long-term stability, as high energy configurations may degrade faster. The weak correlation between power and energy density (0.05) further reinforces the classic energy-power trade-off in electrochemical systems. Overall, the heatmap underscores the critical role of balancing capacitance, power, and stability when designing or predicting optimal configurations.

3.3 ML Algorithms

Machine learning (ML) is becoming an essential tool in material science, helping researchers make faster and more accurate predictions about material properties. This project aims to develop an ML-based model with two key objectives: first, to predict capacitance based on various electrode properties such as specific surface area, pore size, and pore volume; and second, to determine the optimal combination of substrate materials, electrode materials, electrolyte types, and electrolyte concentrations required to achieve a target areal capacitance,energy density or power density.

3.3.1 Machine Learning Models for Capacitance Prediction

A variety of machine learning algorithms were used to assess the performance of the noise-augmented data. These models included Support Vector Regression (SVR), which is effective for high-dimensional data and capturing non-linear relationships due to its foundation in the Support Vector Machine algorithm. The Decision Tree Regressor was also used, leveraging a tree-based structure that splits data based on feature values to predict the target variable. In addition, the Random Forest algorithm, an ensemble technique that builds multiple decision trees and combines their predictions, was applied to improve accuracy and minimize overfitting. XGBoost, a powerful gradient boosting algorithm, was utilized for its ability to combine weak learners into a strong predictive model. Polynomial Regression, which models the relationship between variables as an nth-degree polynomial, was included to capture non-linear patterns. The Multilayer Perceptron (MLP), a type of artificial neural network composed of several layers of interconnected neurons, was also employed to model complex relationships in the data. Beyond these individual models, a stacked ensemble approach was adopted by combining the predictions of the top-performing models.

3.3.2 Machine Learning Models for Material Prediction

For model development, a comparative approach was adopted. Four different regression algorithms were selected based on their proven effectiveness in solving structured data problems: Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and XG Boost Regressor. Each model was integrated into a machine learning pipeline alongside the preprocessing steps. Pipelines ensured that data transformations and model fitting occurred seamlessly and reproducibly without information leakage between training and testing phases. An 80-20 training-test split was employed to evaluate the models, where 80% of the data was used for training, and 20% was held out for testing. In practice, the system was designed to first ask the user which performance metric they'd like to input—whether it's areal capacitance, energy density, or power density. Once the user selected a parameter, they were prompted to enter a target value for that metric. Additionally, the user was given the option to choose whether they wanted to use only pure LIG as the electrode material or if they were open to using doped LIG. If they chose to use doped LIG, the model would then recommend the best possible combination of electrode, electrolyte, substrate, and electrolyte concentration to achieve the desired performance. On the other hand, if they preferred only pure LIG, the model would suggest the most suitable combination of electrolyte, substrate, and electrolyte concentration for that choice. This process is depicted in Figure 4.2. Using an 80-20 training-test split, various regression models—Decision Tree, Random Forest, Gradient Boosting, and XG Boost were tested. Based on their performance on the test set, the most accurate model was selected. Great attention was given to prevent overfitting, a frequent issue in machine learning where models show strong performance on training data but struggle to generalize to unseen data. Techniques such as regularization, and early stopping (particularly for XG Boost) were used to mitigate overfitting risks[33].

3.3.3 Evaluation Metrics

To evaluate the models, multiple performance metrics were employed to measure both predictive accuracy and generalization capability. Among these, the R^2 score was particularly important, as it indicates how much of the variance in the target variable is accounted for by the model. An R^2 value approaching 1 suggests a strong fit, indicating that the model successfully captures the underlying trends within the data. The Root Mean Square Error (RMSE) was also taken into account, as it quantifies the average deviation between predicted and actual values. A lower RMSE value reflects stronger model performance, indicating that the predicted outputs closely align with the actual values on average. This makes it a sensitive metric for evaluating prediction accuracy, especially when large errors are critical to avoid. Another key evaluation metric is the Mean Absolute Error (MAE), which computes the average of the absolute differences between predicted and actual values. Unlike RMSE, MAE treats all errors equally without squaring them, making it less affected by outliers. As with RMSE, a lower MAE indicates that the model's predictions are generally closer to the true values.

3.4 Chapter Summary

The methodology outlines the process of dataset creation, including data collection from literature and augmentation using Gaussian noise. The section also covers the data analysis conducted for both the capacitance prediction and material prediction datasets. It describes the features used for prediction, the regression models employed (SVR, Random Forest, XGBoost, etc.), and evaluation metrics like R^2, MAE, and RMSE.

CHAPTER 4

RESULTS AND DISCUSSION

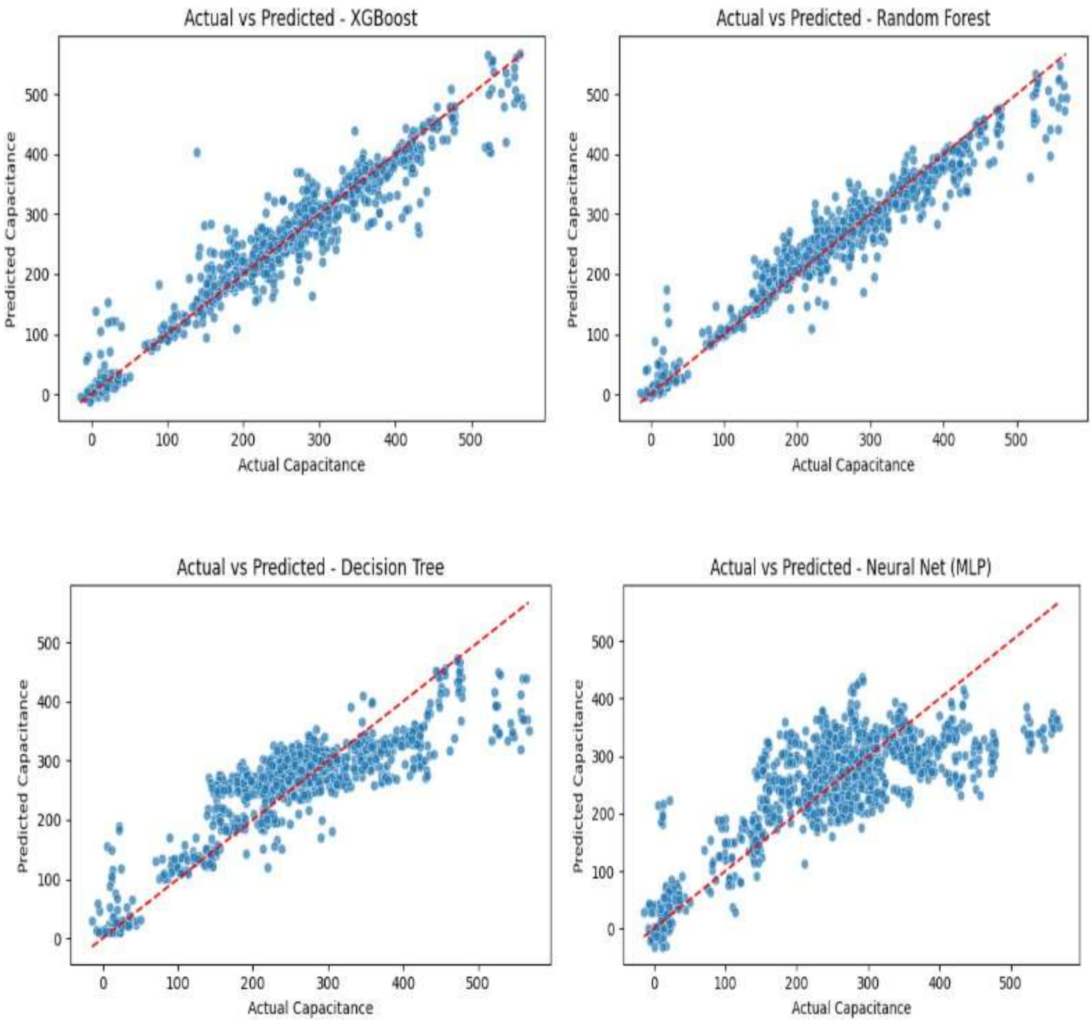
4.1 Capacitance Prediction

4.1.1 Performance Comparison

Table 4.1 provides a comparative evaluation of several regression models used for predicting a target variable, analyzed through key performance metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R^2 Score, and Accuracy percentage. Among all the models tested, the Stacked Model demonstrated superior performance, achieving the lowest RMSE (25.53) and MAE (16.86), alongside the highest R^2 Score (0.95) and Accuracy (95.36%). These values indicate that the Stacked Model offers both high predictive precision and strong correlation with actual values. In this analysis, the Stacked Model refers to a combined approach that leverages the predictive strengths of both Random Forest and XGBoost models. Rather than using these models independently, their predictions are blended typically by averaging to produce a final output. This ensemble approach is designed to exploit the complementary capabilities of both methods. The Random Forest and XGBoost models also performed well, with slightly higher error values but comparable R^2 scores. Their consistent results confirm the effectiveness of ensemble-based techniques in capturing complex relationships in data. In contrast, simpler models like the Decision Tree, Polynomial Linear Regression, Support Vector Regression (SVR), and Neural Networks (MLP) showed significantly higher error values and lower R^2 scores. For instance, the Decision Tree had an RMSE of 53.68 and an R^2 of 0.79, indicating poorer predictive performance. The SVR and Neural Net models, in particular, yielded the lowest accuracy, suggesting that either the models were not optimally tuned or the data's structure did not favor their algorithms.

Table 4.1: Performance comparison of different ML models

Model	**RMSE**	**MAE**	**R^2 Score**	**Accuracy (%)**
Stacked Model	25.53	16.86	0.95	95.36
Random Forest	26.96	17.35	0.95	94.82
XGBoost	29.58	17.98	0.94	93.77
Decision Tree	53.68	39.18	0.79	79.47
Linear Regression (Poly)	67.81	51.32	0.67	67.25
SVR (Tuned)	73.91	50.72	0.61	61.09
Neural Net (MLP)	73.96	56.60	0.61	61.04



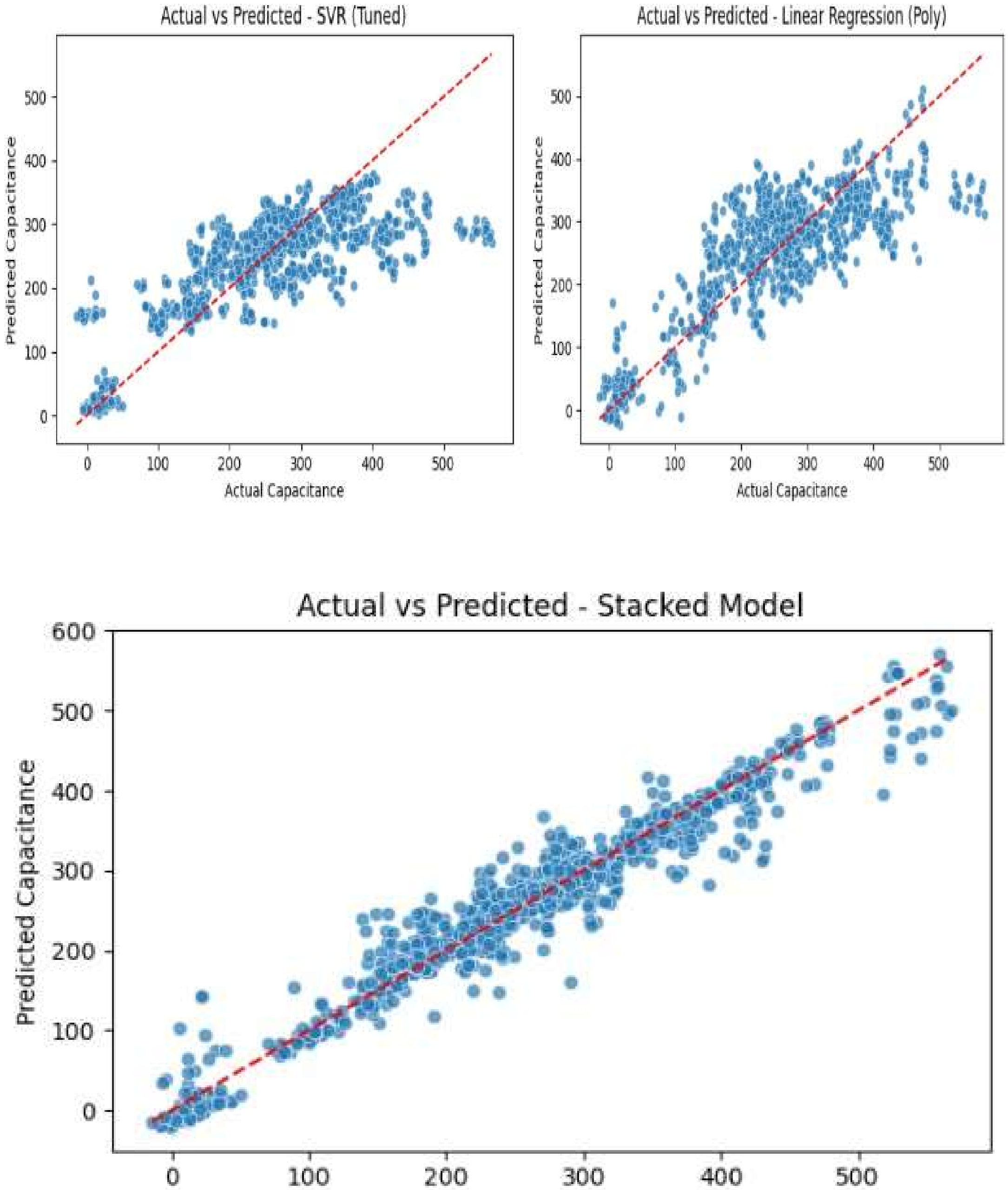


Figure 4.1: Actual vs. Predicted Capacitance for various models: XGBoost, Random Forest, Decision Tree, Neural Net (MLP), SVR (Tuned), Linear Regression (Polynomial), and the final Stacked Model.

The series of scatter plots shown in Figure 4.1 the actual versus predicted capacitance values from seven different regression models applied to the same dataset. These include XGBoost, Random Forest, Decision Tree, Neural Net (MLP), Support Vector Regression (SVR Tuned), Polynomial Linear Regression, and a Stacked Model that combines several base models. Each plot uses a red dashed line to represent the ideal scenario where predicted values match actual values perfectly. Among the individual models, XGBoost and Random Forest exhibit strong predictive performance with points closely following the diagonal, indicating low error and good generalization. The SVR and Neural Net models also show decent accuracy, although with slightly more spread. In contrast, the Decision Tree and Polynomial Linear Regression models demonstrate higher variance and poorer alignment, suggesting overfitting and limitations in capturing complex patterns. The Stacked Model, which integrates the outputs of multiple base learners, clearly outperforms all individual models. It displays the tightest clustering of points around the ideal line, demonstrating that ensembling improves accuracy and robustness by leveraging the strengths of multiple models while mitigating their individual weaknesses.

4.2 Material Prediction

4.2.1 Performance Comparison

This section presents the outcomes of the machine learning modeling phase, in which four regression algorithms—Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor—were applied to a dataset comprising 240 supercapacitor configurations. Each model was trained and evaluated using a standard 80-20 train-test split, allowing for a clear assessment of generalization performance on unseen data. To evaluate model performance, three widely accepted regression metrics were used: Mean Absolute Error (MAE), the Pearson correlation coefficient (R), and the Coefficient of Determination (R^2 score). The Mean Absolute Error (MAE) provides a straightforward measure of the average prediction error, offering an easily interpretable assessment of how closely the model's outputs match the actual values. The correlation coefficient (R) indicates the degree and direction of the

```
Which value do you want to use as input?
1. Areal Capacitance (mF/cm2)
2. Energy Density ( uWh/cm2)
3. Power Density (mW/cm2)
Enter option (1/2/3): 2

You selected: Energy Density ( uWh/cm2)
Allowed range: Min = 0.003, Max = 17.476
Enter your desired value for Energy Density ( uWh/cm2): 0.4

Do you want to use only LIG electrodes? (yes/no): no

📊 Reverse Model Evaluation (80-20 split):
            Model   MAE  R (Correlation)    R²  Accuracy (%)
    Decision Tree 4.823            0.998 0.996        99.61
    Random Forest 4.137            0.998 0.997        99.68
Gradient Boosting 4.375            0.998 0.997        99.67
          XGBoost 4.735            0.998 0.996        99.65

✅ Best model selected: Random Forest

❌ No exact match found in dataset. Proceeding with prediction...

🔮 Predicted material combinations to achieve your target value:
Electrode Electrolyte                     Substrate  Predicted Conc. (g/l)
      LIG   PVA/H2SO4                          Cork                  81.55
    LIG-C   PVA/H2SO4                    parylene-C                  82.57
      LIG   PVA/H2SO4 Whatman chromatography paper                  82.88
      LIG   PVA/H3PO4                     Kapton PI                  86.18
  RuO2/Au   PVA/H2SO4                     Kapton PI                  89.50
```

Figure 4.2: Demonstrating the user interface and interaction flow.

linear relationship between predicted and true values, with values near 1 signifying a strong positive correlation. Meanwhile, the R^2 score represents the fraction of variance in the target variable accounted for by the model, where a value approaching 1 suggests a high-quality fit. These evaluation metrics together provide a comprehensive understanding of each model's predictive ability, highlighting their strengths in minimizing error, maintaining consistency, and effectively capturing underlying data trends. This approach supports the identification of models that can reliably predict optimal material configurations for supercapacitor design, based on the specified target performance criteria.

Figure 4.2 illustrates the execution of the modeling code for an energy density target of 0.4 μWh/cm^2. To determine the optimal material combination for achieving a specified energy density, a reverse modeling approach was applied using four regression algorithms: Decision Tree, Random Forest, Gradient Boosting, and XGBoost. The models were trained to predict the required optimal material combination based on desired performance output. The evaluation of each model was carried out using an 80–20 train-test split to ensure reliability and generalizability. Among the tested algorithms, the Decision Tree Regressor yielded the most favorable performance in terms of predictive accuracy and correlation with actual values. After selecting this model as the most suitable, it was employed to predict the necessary electrolyte concentrations for achieving an energy density of 0.4 μWh/cm^2. Although this target value did not match any exact entry in the dataset, the trained model was able to generate plausible material combinations that closely align with the desired performance. The model's predictions identified several promising combinations. One such configuration included a LIG electrode with a PVA/H_2SO_4 electrolyte and cork substrate, requiring an estimated concentration of 81.55 g/L. Similarly, LIG paired with PVA/H_2SO_4 and Whatman chromatography paper substrate was predicted to require approximately 82.88 g/L. Notably, both traditional and hybrid LIG-based electrodes with various substrates such as Kapton PI and Whatman chromatography paper demonstrated potential to meet the target energy density within similar concentration ranges. These results highlight the usefulness of the reverse modeling framework for supercapacitor optimization. By inputting a desired energy density, the model efficiently narrows down compatible material sets and corresponding electrolyte concentrations. This approach not only accelerates material screening but also supports practical decision-making in experimental design by guiding the synthesis toward data-driven targets. The performance comparison is summarized in Table 4.2.

Table 4.2: Comparison of different regression models

Model	**MAE**	**R Value**	**R^2 Value**	**Accuracy (%)**
Decision Tree	4.823	0.998	0.996	99.61
Random Forest	4.137	0.998	0.997	99.68
Gradient Boosting	4.375	0.998	0.997	99.67
XGBoost	4.735	0.998	0.996	99.65

Among the four models tested, the Random Forest Regressor demonstrated the best performance, achieving a significantly lower Mean Absolute Error (MAE) and a high R^2 score of 0.997. The R^2 value indicates that 99% of the variability in the target variable (e.g., areal capacitance, energy density, or power density) is explained by the input features. This level of performance suggests strong predictive capability and minimal error in the model's predictions. In comparison, the Decision Tree, Gradient Boosting, and XGBoost models, while still effective with R^2 scores of 0.996, 0.997, and 0.996

respectively, exhibited higher errors, suggesting slightly less accurate predictions. The Random Forest's success can be attributed to its ability to model complex, nonlinear relationships while minimizing error. Although it may not offer the same level of flexibility as more advanced models, its high R^2 and accuracy percentage of 99.68% make it particularly suitable for the task at hand, where straightforward and efficient predictions are key. The Gradient Boosting, Decision Tree, and XGBoost models all showed reasonable accuracy but were not able to outperform the Random Forest in terms of MAE or R^2 which is also shown in Figure 5. Notably, XGBoost, despite being a popular choice for many machine learning tasks, performed the worst in this scenario with an R^2 value of 0.996 and accuracy of 99.65%. This outcome may be influenced by various factors such as overfitting, data sparsity, or model configuration issues. Overall, the Random Forest stands out as the optimal model for predictive tasks in this context, as well as for other input variables such as areal capacitance and power density. Its combination of high accuracy and reliable predictions makes it a strong candidate for recommending supercapacitor configurations based on user-defined performance targets.

4.3 Chapter Summary

This chapter presents and compares the performance of different ML models. For capacitance prediction, the Stacked Model outperformed others with an R^2 of 0.95. For material recommendation, the Random Forest Regressor achieved the highest accuracy (99.68%) and lowest error. Detailed performance metrics and visualization plots validate the model predictions and highlight their applicability.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

In this study, we evaluated the performance of various machine learning models for predicting capacitance based on multiple input features. We used a range of performance metrics to assess the models, including R^2 Score, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), with accuracy derived from the R^2 Score. As expected, the models showed a marked improvement in performance as the dataset size increased, particularly in the case of more complex models such as Random Forest Regressor and Gradient Boosting Regressor. The Random Forest Regressor was the standout performer across all dataset sizes, achieving the highest R^2 Score (up to 0.95), which corresponds to 95% accuracy, and the lowest RMSE and MAE values. This demonstrates that Random Forest, with its ability to model complex non-linear relationships in the data, was the most accurate for predicting capacitance. The Decision Tree Regressor and Gradient Boosting Regressor also performed well, especially as the dataset size increased. They showed improvements in R^2 Scores, reaching 92% and 93% accuracy respectively, but they still lagged behind the Random Forest Regressor. The Linear Regression model, while simpler and easier to interpret, performed reasonably well with the larger datasets, achieving an accuracy of 88% with the 1200 dataset. However, it was less effective at handling the more complex, non-linear relationships between the input features and capacitance, which is why it was outperformed by the tree-based models.

For optimal material combination prediction, we conducted a comprehensive evaluation of several machine learning models to predict target performance based on key experimental parameters. The results consistently demonstrated the superior capability of tree-based ensemble models, particularly the Random Forest Regressor, in capturing the complex relationships present within the data. The Random Forest Regressor emerged as the best-performing model, achieving the lowest MAE (4.137) and the highest R^2 Score (0.997), which translates to an accuracy of 99.68%. This high level of precision can be attributed to its ensemble approach, which combines the predictions of multiple decision trees, enabling it to effectively model non-linear interactions between features. Other ensemble models like Gradient Boosting and XGBoost also performed well, with R^2 values of 0.997 and 0.996 respectively, and accuracy exceeding 99.6%, but they exhibited slightly higher error margins than Random Forest. The Decision Tree Regressor, while performing reasonably with a R^2 of 0.996 and accuracy of 99.61%, showed a slightly higher MAE, indicating less stability compared to its ensemble counterparts. Nevertheless, it still proved capable of modeling the data with high reliability. Overall, the results affirm that Random Forest is the most effective and robust model for this prediction task. Its balance of low error and high accuracy makes it especially well-suited for complex regression problems involving multiple interdependent features. While other models demonstrated strong performance, none consistently matched the predictive power and stability of the Random Forest Regressor in this context.

5.2 Future Scope

Looking ahead, one of the key directions for future work is to significantly expand the dataset by incorporating a broader range of experimental data from diverse sources. The current dataset, though sufficient for initial model development, may not fully capture the wide variability seen in real-world supercapacitor systems. By including more variations in material types, fabrication methods, and environmental conditions, the model can be trained to handle a more comprehensive set of scenarios. This would increase its robustness and make it more applicable across a wider range of supercapacitor research and development tasks. Another valuable extension of this work would involve exploring transfer learning or domain adaptation techniques, where the current model is fine-tuned for related tasks using limited new data. This can be particularly useful when applying the model to new electrode materials or configurations that were not included in the original dataset. Ultimately, future efforts will focus on deploying the model in practical, experimental settings to assess its real-world applicability. By validating the model's predictions through actual laboratory tests and performance measurements, its accuracy, reliability, and usefulness can be thoroughly evaluated. These steps will help ensure that the model not only performs well in theory but also serves as a valuable tool for accelerating material discovery and process optimization in the field of energy storage.

5.3 Chapter Summary

The final chapter summarizes the effectiveness of the ML-based framework for super-capacitor optimization. It confirms the suitability of Random Forest for both prediction and recommendation tasks. Future directions include expanding the dataset, integrating transfer learning, and validating predictions through experimental work.

REFERENCES

[1] H. Su, S. Lin, S. Deng, C. Lian, Y. Shang, and H. Liu, "Predicting the capacitance of carbon-based electric double layer capacitors by machine learning," *Nanoscale Advances*, vol. 1, no. 6, pp. 2162–2166, 2019.

[2] K. Subasinghage, K. Gunawardane, N. Padmawansa, N. Kularatna, and M. Moradian, "Modern supercapacitors technologies and their applicability in mature electrical engineering applications," *Energies*, vol. 15, no. 20, p. 7752, 2022.

[3] M. C. A. de Oliveira, A. L. Freire, I. C. Candido, Y. Messaddeq, J. Lapointe, J. Fréchette, R. Vallée, and H. P. de Oliveira, "Flexible micro supercapacitor and triboelectric nanogenerator based on laser-induced graphene for self-powered applications," *Next Energy*, vol. 4, p. 100141, 2024.

[4] K. O. Oyedotun and B. B. Mamba, "New trends in supercapacitors applications," *Inorganic Chemistry Communications*, p. 113154, 2024.

[5] S. Ahmed, M. Rafat, and S. Hashmi, "Effect of electrolyte concentration on supercapacitor performance of graphene-nio composite," *Int. J. Adv. Res. Sci. Eng.*, vol. 6, pp. 609–614, 2017.

[6] A. González, E. Goikolea, J. A. Barrena, and R. Mysyk, "Review on supercapacitors: Technologies and materials," *Renewable and sustainable energy reviews*, vol. 58, pp. 1189–1206, 2016.

[7] D. P. Chatterjee and A. K. Nandi, "A review on the recent advances in hybrid supercapacitors," *Journal of Materials Chemistry A*, vol. 9, no. 29, pp. 15 880–15 918, 2021.

[8] M. Horn, J. MacLeod, M. Liu, J. Webb, and N. Motta, "Supercapacitors: A new source of power for electric cars?" *Economic Analysis and Policy*, vol. 61, pp. 93–103, 2019.

[9] A. Berrueta, A. Ursua, I. San Martin, A. Eftekhari, and P. Sanchis, "Supercapacitors: electrical characteristics, modeling, applications, and future trends," *Ieee Access*, vol. 7, pp. 50 869–50 896, 2019.

[10] M. Khodaparastan and A. Mohamed, "Supercapacitors for electric rail transit systems," in *2017 IEEE 6th International Conference on Renewable Energy Research and Applications (ICRERA)*. IEEE, 2017, pp. 896–901.

[11] X.-Y. Fu, R.-Y. Shu, C.-J. Ma, Y.-Y. Zhang, H.-B. Jiang, and M.-N. Yao, "Self-assembled mxene-graphene oxide composite enhanced laser-induced graphene based electrodes towards conformal supercapacitor applications," *Applied Surface Science*, vol. 631, p. 157549, 2023.

[12] X.-Y. Fu, R.-Y. Shu, C.-J. Ma, H.-B. Jiang, and M.-N. Yao, "Graphene oxide assisted-mno2 nanoparticles enhanced laser-induced graphene based electrodes for supercapacitor application," *Electrochimica Acta*, vol. 481, p. 143987, 2024.

[13] M. Z. Iqbal, S. Zakar, and S. S. Haider, "Role of aqueous electrolytes on the performance of electrochemical energy storage device," *Journal of Electroanalytical Chemistry*, vol. 858, p. 113793, 2020.

[14] A. Acharjee and B. Saha, "Organic electrolytes in electrochemical supercapacitors: Applications and developments," *Journal of Molecular Liquids*, p. 124487, 2024.

[15] P. Mohapatra and A. K. Barick, "Ionic liquids based polymer electrolytes for supercapacitor applications," *Journal of Power Sources*, vol. 626, p. 235749, 2025.

[16] X. Cai, Y. Xiao, and W. Sun, "Pore-size effect of activated carbons on the electrochemical performances of symmetric supercapacitors under compression," *Journal of Energy Storage*, vol. 55, p. 105438, 2022.

[17] J. Schmidt, M. R. Marques, S. Botti, and M. A. Marques, "Recent advances and applications of machine learning in solid-state materials science," *npj computational materials*, vol. 5, no. 1, p. 83, 2019.

[18] Y. Liu, T. Zhao, W. Ju, and S. Shi, "Materials discovery and design using machine learning," *Journal of Materiomics*, vol. 3, no. 3, pp. 159–177, 2017.

[19] C. Wang, X. P. Tan, S. B. Tor, and C. Lim, "Machine learning in additive manufacturing: State-of-the-art and perspectives," *Additive Manufacturing*, vol. 36, p. 101538, 2020.

[20] E. Westphal and H. Seitz, "A machine learning method for defect detection and visualization in selective laser sintering based on convolutional neural networks," *Additive Manufacturing*, vol. 41, p. 101965, 2021.

[21] A. E. Bilali, A. Taleb, M. A. Bahlaoui, and Y. Brouziyne, "An integrated approach based on gaussian noises-based data augmentation method and adaboost model to predict faecal coliforms in rivers with small dataset," *Journal of Hydrology*, vol. 599, p. 126510, 2021.

[22] K. Wang, L. Li, Y. Li, L. Tao, L. Wang, S. Yuan, H. Fu, C. Geng, and X. Xu, "Preparation of graphene carbon nanotube supercapacitor electrode materials for new energy vehicles," *Materials Technology*, vol. 39, no. 1, p. 2372945, 2024.

[23] G. D'Altri, L. Yeasmin, V. Di Matteo, S. Scurti, A. Giovagnoli, M. F. Di Filippo, I. Gualandi, M. C. Cassani, D. Caretti, S. Panzavolta *et al.*, "Preparation and characterization of self-healing pva–h2so4 hydrogel for flexible energy storage," *ACS omega*, vol. 9, no. 6, pp. 6391–6402, 2024.

[24] H. Tariq, S. U. Awan, D. Hussain, S. Rizwan, S. A. Shah, S. Zainab, and M. B. Riaz, "Enhancing supercapacitor performance through design optimization of laser-induced graphene and mwcnt coatings for flexible and portable energy storage," *Scientific Reports*, vol. 13, no. 1, p. 21116, 2023.

[25] R. Correia, J. Deuermeier, M. R. Correia, J. Vaz Pinto, J. Coelho, E. Fortunato, and R. Martins, "Biocompatible parylene-c laser-induced graphene electrodes for microsupercapacitor applications," *ACS applied materials & interfaces*, vol. 14, no. 41, pp. 46 427–46 438, 2022.

[26] Z. Peng, R. Ye, J. A. Mann, D. Zakhidov, Y. Li, P. R. Smalley, J. Lin, and J. M. Tour, "Flexible boron-doped laser-induced graphene microsupercapacitors," *ACS nano*, vol. 9, no. 6, pp. 5868–5875, 2015.

[27] K. Brousse, S. Pinaud, S. Nguyen, P.-F. Fazzini, R. Makarem, C. Josse, Y. Thimont, B. Chaudret, P.-L. Taberna, M. Respaud *et al.*, "Facile and scalable preparation of ruthenium oxide-based flexible micro-supercapacitors," *Advanced Energy Materials*, vol. 10, no. 6, p. 1903136, 2020.

[28] G. K. Maron, M. G. Masteghin, V. Gehrke, L. S. Rodrigues, J. H. Alano, J. H. Rossato, V. R. Mastelaro, J. Dupont, M. T. Escote, S. R. P. Silva *et al.*, "Enhanced pseudocapacitive behaviour in laser-written graphene micro-supercapacitors decorated with nickel cobalt sulphide nanoparticles," *Materials research bulletin*, vol. 168, p. 112490, 2023.

[29] C. Ryu, H. M. Do, and J. B. In, "Enhanced performance of densified laser-induced graphene supercapacitor electrodes in dimpled polyimide," *Applied Surface Science*, vol. 643, p. 158696, 2024.

[30] J. Coelho, R. F. Correia, S. Silvestre, T. Pinheiro, A. C. Marques, M. R. P. Correia, J. V. Pinto, E. Fortunato, and R. Martins, "based laser-induced graphene for sustainable and flexible microsupercapacitor applications," *Microchimica Acta*, vol. 190, no. 1, p. 40, 2023.

[31] F. Mahmood, H. Zhang, J. Lin, and C. Wan, "Laser-induced graphene derived from kraft lignin for flexible supercapacitors," *ACS omega*, vol. 5, no. 24, pp. 14 611–14 618, 2020.

[32] S. Silvestre, T. Pinheiro, A. Marques, J. Deuermeier, J. Coelho, R. Martins, L. Pereira, and E. Fortunato, "Cork derived laser-induced graphene for sustainable green electronics. flex print electron 7: 035021," 2022.

[33] Y. Tian and Y. Zhang, "A comprehensive survey on regularization strategies in machine learning," *Information Fusion*, vol. 80, pp. 146–166, 2022.

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